

DATA ANALYTICS

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FINAL EXAMINATION & TUTORIAL QUESTION BANK — WITH ANSWERS

PART – A: MANDATORY QUESTIONS

Q1(a) — Define Data Analytics. Explain how data analytics can be converted into a business insight. Use a real-life Bangladesh example. [3 Marks]

Data Analytics is the process of examining raw data using statistical and computational methods to discover patterns, draw conclusions, and generate meaningful, actionable insights that support decision-making. It is a form of Business Intelligence that transforms raw numbers into knowledge.

The conversion process follows these stages:

- Raw Data Collection — Gathering data from transactions, sensors, or user activity.
- Data Cleaning & Processing — Removing errors and standardizing the data.
- Analysis — Applying statistical tools to identify patterns and trends.
- Insight Generation — Translating patterns into a clear business finding.
- Decision & Action — Using the finding to make smarter business choices.

Real-life Bangladesh Example — Bkash: Bkash collects millions of mobile financial transaction records daily. By analyzing when and where transactions fail, which regions show sudden drops in activity, and which user segments are most active, Bkash's analysts can identify fraud patterns early, optimize server capacity during Eid payment surges, and create targeted promotions for inactive users — all replacing guesswork with data-driven decisions.

Q1(b) — Regression Equation: $Y = 40 + 6X$. (i) Predict sales when advertising = 8000 BDT. (ii) Interpret slope and intercept. [3 Marks]

(i) Prediction when $X = 8$ (thousands BDT)

$$Y = 40 + 6X$$

$$Y = 40 + 6(8) = 40 + 48 = 88 \text{ units}$$

Answer: When advertising expenditure is 8,000 BDT, predicted sales = 88 units.

(ii) Interpretation

Intercept (40): When zero BDT is spent on advertising, the model predicts 40 units will still be sold. This represents the baseline sales achieved through brand reputation, word-of-mouth, or existing customer loyalty without any advertising effort.

Slope (6): For every additional 1,000 BDT spent on advertising, sales are expected to increase by 6 units. This means advertising has a positive and consistent impact on sales volume.

Q1(c) — Find Median, Q1, Q3, and IQR for: {14, 18, 11, 13, 6, 8, 2, 5, 19, 21, 23, 20, 23, 27, 25, 24, 31} [3 Marks]

Step 1: Arrange in Ascending Order

2, 5, 6, 8, 11, 13, 14, 18, 19, 20, 21, 23, 23, 24, 25, 27, 31

$n = 17$

Step 2: Median (Q2)

Median position = $(n+1)/2 = (17+1)/2 = 9\text{th term}$

9th term = 19

Median = 19

Step 3: Lower Quartile (Q1)

Q1 position = $(n+1)/4 = (17+1)/4 = 4.5\text{th term}$

Average of 4th and 5th terms = $(8+11)/2 = 9.5$

Q1 = 9.5

Step 4: Upper Quartile (Q3)

Q3 position = $3(n+1)/4 = 3(18)/4 = 13.5\text{th term}$

Average of 13th and 14th terms = $(23+24)/2 = 23.5$

Q3 = 23.5

Step 5: IQR

$IQR = Q3 - Q1 = 23.5 - 9.5 = 14$

Final Answers: Median = 19, Q1 = 9.5, Q3 = 23.5, IQR = 14

Q1(d) — How do outliers affect statistical analysis and ML models? What techniques determine whether to eliminate or keep them? [3 Marks]

Effects of Outliers

- Mean: Severely distorted — a single extreme value can pull the mean far from the typical value. The median is more robust.
- Standard Deviation & Variance: Outliers inflate these measures, making data appear far more spread out than it truly is.
- Regression Models: Outliers shift the regression line, producing misleading slope/intercept values and poor predictions.
- Machine Learning Models: In distance-based models (KNN, K-Means), outliers create wrong cluster assignments. In tree-based models, they can create unnecessary splits. Neural networks may overfit to outlier patterns.

Techniques to Decide Whether to Keep or Remove Outliers

- Z-Score Method: If a value's Z-score is beyond ± 3 , it is a statistical outlier. Only remove it if it cannot represent a real event.
 - IQR Method: Values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ are outliers. Investigate the cause before removal.
 - Box Plot Visualization: Points plotted beyond whiskers are outliers — visual inspection helps decide.
 - Domain Knowledge Check: If the outlier is physically/logically impossible (e.g., age = 300), remove it. If it is extreme but real (e.g., a billionaire's income in a salary dataset), keep it or treat it separately.
 - Winsorization: Replace extreme values with the nearest acceptable boundary rather than deleting them.
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Q1(e) — How to deal with missing values in a mixed dataset (PQS.csv)? THREE tactics and their effect on model performance. [3 Marks]

Tactic 1: Deletion (Listwise or Pairwise)

Remove rows with missing values entirely (listwise) or exclude them from specific calculations (pairwise).

Effect: Simple and prevents bias IF the data is Missing Completely At Random (MCAR). However, if many rows are deleted, the reduced dataset may cause the model to underfit and lose generalization power. For small datasets, this is particularly damaging.

Tactic 2: Imputation

Fill missing numerical values with the mean, median, or mode of that column. For categorical attributes, use the most frequent category.

Effect: Preserves dataset size and allows the model to train on more data. However, mean imputation reduces variance and can distort relationships between variables, potentially weakening model accuracy. Median imputation is safer when data is skewed.

Tactic 3: Predictive Imputation / Model-Based Filling

Use a regression model or K-Nearest Neighbor algorithm to predict and fill missing values based on other attributes in the same row.

Effect: Produces more accurate fill values that reflect real data patterns, leading to better model performance compared to simple mean imputation. However, it is computationally expensive and can introduce its own prediction errors, especially if the imputation model itself is inaccurate.

Q1(f) — KNN Algorithm: (i) Effect of K=1 vs K=5. (ii) Which K is more robust against noise? [3 Marks]

(i) Effect of K Value on Classification

K = 1: The new data point is classified based on its single nearest neighbor. This is extremely sensitive — if that one neighbor is a noisy or mislabeled point, the classification will be wrong. The decision boundary is highly irregular and complex, leading to overfitting to training data.

K = 5: The new point is classified by majority vote among 5 nearest neighbors. This smooths out the decision boundary. Even if 1 or 2 neighbors are noisy, the majority vote still produces a sensible classification. This leads to better generalization on unseen data.

(ii) Which K is More Robust Against Noise?

K = 5 is significantly more robust against noise. With majority voting across 5 neighbors, a single mislabeled or extreme point cannot override the classification. The larger K averages out local irregularities.

However, K must not be too large either — a very high K makes the model overly simple and may miss local patterns (underfitting). The optimal K is typically found using cross-validation.

Q1(g) — Distinguish Descriptive vs Predictive Analytics. Should Prescriptive come after both? [2 Marks]

Dimension	Descriptive Analytics	Predictive Analytics
Key Question	What happened?	What will happen?
Purpose	Summarize historical data	Forecast future outcomes
Techniques	Aggregation, Data Mining	Regression, ML models
Example	Monthly sales report	Predicting customer churn

Complexity	Low	Medium-High
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Should Prescriptive Analytics Follow a Hierarchy?

YES — following the hierarchy Descriptive → Predictive → Prescriptive is a logically sound and practically beneficial approach. Here is why:

- Descriptive analytics provides the foundation by revealing what has already happened. Without this understanding, building predictive models on unknown patterns is unreliable.
- Predictive analytics uses the patterns found in descriptive analysis to forecast what is likely to happen next. It depends on clean, well-understood historical data.
- Prescriptive analytics recommends the best course of action. It relies on predictions from predictive analytics to evaluate scenarios and suggest optimal decisions.

Skipping descriptive or predictive analytics before applying prescriptive analytics would mean recommending actions without understanding the current situation or likely future — like prescribing medicine without a diagnosis. The hierarchy ensures decisions are evidence-based at every level.

PART – B: QUESTIONS 2–7

Question 2

Q2(a) — Define Big Data. Explain the 4 Vs with real-world examples. [3 Marks]

Big Data refers to extremely large, fast-moving, and varied datasets that cannot be effectively processed, stored, or analyzed using traditional data management tools. It is defined by four key characteristics called the 4 Vs:

V	Definition	Real-World Example
Volume	Massive quantity of data generated	Facebook generates over 4 petabytes of data per day from posts, likes, and messages
Velocity	Speed at which data is generated and processed	Twitter processes 500 million tweets per day — data must be analyzed in near real-time
Variety	Data comes in many formats: text, image, video, sensor data	A hospital collects X-ray images (unstructured), patient records (structured), and doctor notes (semi-structured)
Veracity	Uncertainty and accuracy of data; some data may be noisy or unreliable	Social media sentiment data may contain sarcasm, slang, or fake accounts that reduce accuracy

Q2(b) — Structured vs Unstructured Data. TWO examples each. How are they stored and processed differently? [3 Marks]

Dimension	Structured Data	Unstructured Data
Definition	Organized in rows and columns with a defined schema	No predefined format or structure
Example 1	SQL database table of customer orders	Email messages and attachments
Example 2	Excel spreadsheet of monthly sales	Video recordings and social media images

Storage	Relational databases (MySQL, PostgreSQL)	NoSQL databases (MongoDB), file systems, data lakes
Processing	SQL queries, OLAP cubes, spreadsheet tools	NLP for text, computer vision for images, Hadoop/Spark for large volumes

Q2(c) — Relationship between Business Strategy and Business Analytics. [3 Marks]

Business strategy defines WHERE an organization wants to go and WHAT it aims to achieve. Business Analytics provides the HOW — by using data to evaluate options, measure progress, and make evidence-based decisions.

- Analytics enables strategy by quantifying market opportunities, customer behavior, and operational performance, giving leaders concrete data instead of gut feeling.
- Analytics measures strategy by tracking KPIs (Key Performance Indicators) against strategic goals in real time using dashboards and scorecards.
- Analytics refines strategy by uncovering patterns that reveal when strategies are underperforming, allowing rapid adjustment.

Example: A telecom company's strategy is to reduce customer churn. Business analytics identifies which customers are most likely to leave (predictive analytics), why they are leaving (diagnostic analytics), and what offer would retain them (prescriptive analytics). Analytics transforms the strategy from a vague goal into a targeted, executable plan.

Q2(d) — Monthly Sales Line Graph for Company X and Company Y. [3 Marks]

Data Table:

Particulars	Jan	Feb	Mar	Apr	May	Jun
Company X	140	135	140	155	170	189
Company Y	220	215	210	205	190	172

Sales Trend Commentary:

- Company X shows a consistent upward trend, starting at 140K BDT in January and rising to 189K BDT by June — an overall increase of approximately 35%. The growth accelerates from April onward, suggesting a successful mid-year strategy or seasonal demand.
- Company Y shows a steady declining trend, starting at 220K BDT and falling to 172K BDT by June — a decrease of approximately 22%. This suggests increasing competition, loss of market share, or product saturation.
- The two companies' sales converge over time — Company X may overtake Company Y if current trends continue into H2 2023.

Q2(e) — Define Analytics Ecosystem. Name FIVE key players and their roles. [2 Marks]

The Analytics Ecosystem is the full network of organizations, people, and tools that work together to enable data-driven decision-making around a central analytics user organization.

Player	Role
Analytics User Organization	The business at the center that uses analytics to make decisions
Data Generation Infrastructure Providers	Supply sensors, apps, and platforms that create raw data (e.g., IoT device manufacturers)

Data Warehouse Providers	Store large volumes of cleaned, organized data ready for analysis (e.g., Amazon Redshift)
Analytics-Focused Software Developers	Build tools for querying, visualizing, and analyzing data (e.g., Tableau, Power BI)
Regulators and Policy Makers	Create laws governing how data can be collected, stored, and used (e.g., GDPR in Europe)
Academic Institutions & Certifiers	Train data professionals and set standards for analytics knowledge and credentials

Question 3

Q3(a) — Machine Learning for Business Analytics. Supervised vs Unsupervised Learning. [4 Marks]

Machine learning (ML) enables computers to learn from historical business data and make predictions or decisions without being explicitly programmed for each scenario. In business analytics:

- Predictive modeling uses ML to forecast sales, detect fraud, and predict customer churn.
- Customer segmentation uses ML to group customers by behavior for targeted marketing.
- Recommendation systems suggest products based on past purchases (e.g., Daraz, Amazon).
- Anomaly detection identifies unusual patterns in financial transactions or network traffic.

Dimension	Supervised Learning	Unsupervised Learning
Definition	Model learns from labeled training data (input-output pairs)	Model finds hidden patterns in unlabeled data on its own
Goal	Predict a known output category or value	Discover structure or groupings in the data
Techniques	Classification, Regression, KNN, Decision Trees	K-Means Clustering, Hierarchical Clustering, PCA
Example	Email spam filter — each email is labeled 'spam' or 'not spam'	Customer segmentation — grouping customers by buying behavior without predefined groups
Output	Specific prediction (class or number)	Clusters, patterns, or reduced dimensions

Q3(b) — Find Mean and Median from grouped overtime distribution. [4 Marks]

Distribution:

Overtime (hrs.)	10–15	15–20	20–25	25–30	30–35
No. of Employees	11	20	35	20	18

Mean Calculation (Using Midpoint Method):

Class	Midpoint (m)	Freq (f)	f × m
10–15	12.5	11	137.5
15–20	17.5	20	350

20–25	22.5	35	787.5
25–30	27.5	20	550
30–35	32.5	18	585
Total	—	100	2410

Mean = $\Sigma(f \times m) / \Sigma f = 2410 / 100 = 24.1$ hours

Median Calculation:

N = 100, N/2 = 50. Cumulative Frequencies: 10–15: 11 | 10–20: 31 | 10–25: 66

The 50th value falls in the 20–25 class (cumulative reaches 66 for first time, after 31).

Median = $L + [(N/2 - F) / f] \times h = 20 + [(50 - 31) / 35] \times 5 = 20 + (19/35) \times 5 = 20 + 2.71 = 22.71$ hours

Mean = 24.1 hours, Median = 22.71 hours

Q3(c) — Sensors, apps, IoT devices — type/category of data they generate. [3 Marks]

Device/System	Data Generated	Type/Category	Justification
Temperature Sensor (IoT)	Numeric temperature readings every second	Quantitative, Structured, Streaming	Numbers in a defined format, generated continuously at high velocity
Fitness Tracker App	Steps, heart rate, GPS location	Quantitative + Semi-structured	Numeric values with timestamps; GPS adds spatial dimensions
CCTV Camera	Video footage	Unstructured, Multimedia	No predefined schema; requires computer vision for processing
Social Media App	Text posts, images, likes, comments	Unstructured, Qualitative + Quantitative mixed	Variety of formats; text is qualitative, likes are quantitative
Smart Electricity Meter	Power consumption readings at intervals	Quantitative, Structured, Time-Series	Regular numeric readings in fixed format — classic structured IoT data

Q3(d) — Dataset {2, 5, 8, 12, 15, 18, 21, 25}: (i) Standard Deviation & Variance. (ii) Range & Coefficient of Range. [3 Marks]

(i) Standard Deviation and Variance

Data: 2, 5, 8, 12, 15, 18, 21, 25 n = 8

Step 1: Mean = $(2+5+8+12+15+18+21+25) / 8 = 106 / 8 = 13.25$

Step 2: $(X - \text{Mean})^2$:

$(2-13.25)^2=126.5625$, $(5-13.25)^2=68.0625$, $(8-13.25)^2=27.5625$, $(12-13.25)^2=1.5625$

$(15-13.25)^2=3.0625$, $(18-13.25)^2=22.5625$, $(21-13.25)^2=60.0625$, $(25-13.25)^2=138.0625$

Sum = $126.5625+68.0625+27.5625+1.5625+3.0625+22.5625+60.0625+138.0625 = 447.5$

Step 3: Variance = $447.5 / 8 = 55.9375 \approx 55.94$

Step 4: Standard Deviation = $\sqrt{55.94} \approx 7.48$

Variance = 55.94, Standard Deviation = 7.48

(ii) Range and Coefficient of Range

X_max = 25, X_min = 2

Range = $25 - 2 = 23$

Coefficient of Range = $(X_{\text{max}} - X_{\text{min}}) / (X_{\text{max}} + X_{\text{min}}) = (25-2) / (25+2) = 23 / 27 \approx 0.852$
Range = 23, Coefficient of Range ≈ 0.852

Question 4

Q4(a) — What is classification? Differences between classification and clustering. [3 Marks]

Classification is a supervised machine learning task where the algorithm learns from labeled training data to assign new data points to one of a set of predefined categories or classes.

Dimension	Classification	Clustering
Type	Supervised Learning	Unsupervised Learning
Training Data	Requires labeled data (class known)	No labels required — finds groups automatically
Goal	Predict which predefined class a new item belongs to	Discover natural groupings or segments in data
Output	Class label (e.g., Spam/Not Spam)	Clusters or segments (e.g., Customer Group A, B, C)
Example	KNN classifying emails as spam or not spam	K-Means grouping customers by purchasing behavior
Evaluation	Accuracy, Precision, Recall, F1-Score	Silhouette Score, Elbow Method

Q4(b) — Find availability for Systems P and Q. Which is more reliable? [3 Marks]

Availability = $MTTF / (MTTF + MTTR)$

System P:

MTTF = 900 hrs, MTTR = 100 hrs

Availability = $900 / (900 + 100) = 900 / 1000 = 0.90 = 90\%$

System Q:

MTTF = 500 hrs, MTTR = 25 hrs

Availability = $500 / (500 + 25) = 500 / 525 \approx 0.952 = 95.2\%$

Answer: System Q (95.2%) is more reliable in terms of availability than System P (90%), despite having a lower MTTF. This is because System Q recovers from failures much faster (MTTR = 25 hrs vs 100 hrs), resulting in less total downtime.

Q4(c) — KNN (K=5) to classify new patient (Age=38, BP=135). [4 Marks]

Dataset:

Age	Blood Pressure	Risk Level
25	120	Low
35	130	Low
45	150	High
50	160	High
30	125	Low
40	145	High

Step 1: Calculate Euclidean Distance from (38, 135) to each point:

$$d = \sqrt{(\text{Age} - 38)^2 + (\text{BP} - 135)^2}$$

Point	Age	BP	Distance Calculation	Distance	Class
P1	25	120	$\sqrt{[(25-38)^2 + (120-135)^2]} = \sqrt{169+225}$	$\sqrt{394} \approx 19.85$	Low
P2	35	130	$\sqrt{[(35-38)^2 + (130-135)^2]} = \sqrt{9+25}$	$\sqrt{34} \approx 5.83$	Low
P3	45	150	$\sqrt{[(45-38)^2 + (150-135)^2]} = \sqrt{49+225}$	$\sqrt{274} \approx 16.55$	High
P4	50	160	$\sqrt{[(50-38)^2 + (160-135)^2]} = \sqrt{144+625}$	$\sqrt{769} \approx 27.73$	High
P5	30	125	$\sqrt{[(30-38)^2 + (125-135)^2]} = \sqrt{64+100}$	$\sqrt{164} \approx 12.81$	Low
P6	40	145	$\sqrt{[(40-38)^2 + (145-135)^2]} = \sqrt{4+100}$	$\sqrt{104} \approx 10.20$	High

Step 2: Select K=5 Nearest Neighbors (sorted by distance):

P2 (5.83, Low), P6 (10.20, High), P5 (12.81, Low), P3 (16.55, High), P1 (19.85, Low)

Step 3: Majority Vote:

Low: P2, P5, P1 = 3 votes | High: P6, P3 = 2 votes

Classification Result: LOW RISK — The new patient (Age=38, BP=135) is classified as Low Risk based on K=5 nearest neighbor majority vote.

Q4(d) — Reliability metrics. Lazy learner vs Eager learner. [2 Marks]**Reliability Metrics:**

- MTTF (Mean Time To Failure) — average time a system operates before failing
- MTTR (Mean Time To Repair) — average time to restore after a failure
- Availability = $\text{MTTF} / (\text{MTTF} + \text{MTTR})$
- Reliability = $e^{(-\lambda t)}$ where λ = failure rate

Dimension	Lazy Learner	Eager Learner
Training	Stores training data; no model built at train time	Builds a generalized model during training
Prediction	Computes prediction at query time using stored data	Uses pre-built model for fast predictions
Speed	Slow at prediction (calculates distances on the fly)	Fast at prediction
Example	K-Nearest Neighbor (KNN)	Decision Tree, Neural Network, Logistic Regression
Memory	High — stores entire training dataset	Low — only stores the model parameters

Question 5

Q5(a) — Explain the 7-step Data Analysis Process. One tool per step. [4 Marks]

Step	Name	Description	Tool/Technique
1	Define the Question	Clearly identify the business problem and form a specific, testable hypothesis	KPI Dashboards (Grafana, Databox)
2	Collect the Data	Gather relevant data — first-party (own), second-party (partner), or third-party (public)	Salesforce DMP, Kaggle, data.gov
3	Clean the Data	Remove errors, duplicates, outliers; standardize formats; handle missing values	Python (Pandas), OpenRefine, Data Ladder
4	Analyze the Data	Apply descriptive, diagnostic, predictive, or prescriptive techniques to find patterns	Python, R, Excel, SPSS, Tableau
5	Share Results	Present findings visually and clearly to stakeholders and decision-makers	Tableau, Google Charts, Matplotlib, Seaborn
6	Embrace Failure	If results are unclear, go back, revise the question, collect more data, or re-clean	Iterative workflow tools, version control
7	Continuous Improvement	Apply lessons learned to improve methodology in the next analytics project	Process documentation, retrospectives

Q5(b) — Define Data Quality. SIX dimensions with examples. Why is it critical for ML? [3 Marks]

Data Quality measures how well a dataset is suited for its intended purpose. High-quality data is accurate, complete, and reliable for use in analysis and machine learning.

Dimension	Definition	Example
Accuracy	Data correctly represents the real-world entity	A customer's phone number must be a real, dialable number
Completeness	All required data fields are present	A contacts list must have names for all entries if alphabetical sorting is the goal
Consistency	Data matches across all systems and sources	A patient cannot have two different dates of birth in two hospital systems
Validity	Data conforms to defined formats, ranges, and rules	A birth year of 1700 is invalid — outside the realistic human lifespan range
Uniformity	All values use the same unit, format, or scale	Weight data must be all in kg or all in lbs, not mixed
Relevance (Timeliness)	Data is current and applicable to the task at hand	A board meeting needs this quarter's profits, not last year's figures

Why Critical for ML: Machine learning models learn patterns from training data. Poor quality data leads to models learning incorrect patterns ('garbage in, garbage out'). Inaccurate data produces wrong predictions; incomplete data causes models to miss important features; inconsistent data creates

confusion in the learning algorithm. Even the most advanced ML algorithm cannot compensate for fundamentally flawed data.

Q5(c) — Evolution of Decision Support Systems from 1970s to Big Data era. How did BI emerge? [3 Marks]

Era	Technology	Key Innovation
1970s	Decision Support Systems (DSS), Expert Systems	First computer-aided systems to help managers make structured decisions
1980s	Relational DBMS, ERP Systems, On-Demand Reporting	Databases allowed structured storage; ERPs integrated business operations
1990s	Data Warehousing, Dashboards, Executive Information Systems (EIS)	Separate analytical data stores emerged; executives could view summaries
2000s	Business Intelligence (BI), Data Mining, SaaS, Cloud Computing	BI platforms consolidated analytics; cloud made tools accessible company-wide
2010s	Big Data, Hadoop, NoSQL, Social Media Analytics	Explosion of unstructured data from mobile and internet required new processing paradigms

Business Intelligence emerged in the 2000s as organizations needed a unified way to collect data from multiple sources (ERP, CRM, databases), store it in data warehouses, and present insights through dashboards and reports. BI grew directly from the limitations of earlier DSS — where DSS helped individuals make specific decisions, BI gave entire organizations real-time, self-service access to operational and strategic data.

Q5(d) — Linear Regression error metrics (RMSE and MAE). [3 Marks]

Actual (x)	100	150	200	250	300
Predicted (x_p)	110	145	195	260	290

Error Calculations:

x	x_p	Error (x-x_p)	Error	Error ²
100	110	-10	10	100
150	145	5	5	25
200	195	5	5	25
250	260	-10	10	100
300	290	10	10	100
Total	—	—	40	350

$MAE = \Sigma |Error| / n = 40 / 5 = 8$
 $MSE = \Sigma Error^2 / n = 350 / 5 = 70$
 $RMSE = \sqrt{MSE} = \sqrt{70} \approx 8.37$

MAE = 8.0, RMSE ≈ 8.37

Interpretation: On average, the model's predictions are off by 8 units (MAE). The RMSE of 8.37 penalizes larger errors more heavily, indicating occasional larger prediction errors.

Question 6

Q6(a) — What is Business Intelligence? Describe BI Architecture. [4 Marks]

Business Intelligence (BI) is a broad set of technologies, processes, and tools that collect, store, integrate, and analyze business data to support informed decision-making. The fundamental goal is to transform raw operational data into actionable insights for managers and executives.

The BI flow: Data → Information → Decision → Action

BI Architecture Components:

Component	Role
Data Sources	The origin of all data — operational databases, ERP systems, CRM systems, spreadsheets, and external APIs. Multiple heterogeneous sources feed into the BI system.
ETL (Extract, Transform, Load)	Extract: Pull raw data from source systems. Transform: Clean, reformat, and standardize data. Load: Insert processed data into the data warehouse. ETL ensures data quality and consistency.
Data Warehouse	A centralized, subject-oriented repository that stores cleaned, historical, and integrated data optimized for analysis — not for day-to-day transactions.
Data Marts	Smaller, department-specific subsets of the data warehouse (e.g., a Finance Data Mart, a Sales Data Mart) that serve the specific analytical needs of individual business units.
Analytical Tools	OLAP cubes for multidimensional analysis, dashboards for KPI monitoring, data mining for pattern discovery, and reporting tools for structured summaries.
User Interface	Browsers, portals, and dashboards that present insights to business users in an accessible, visual format — no technical knowledge required to read.

Q6(b) — How can Predictive Analytics enhance resource allocation and ICT service delivery? TWO use cases. [2 Marks]

Use Case 1: Network Capacity Planning (Telecom/ISP)

Predictive analytics models analyze historical internet usage patterns — by time of day, location, and user segment. Providers can predict peak load times and allocate additional server bandwidth before congestion occurs, improving service quality and reducing costly emergency infrastructure upgrades.

Use Case 2: Hospital Staff Scheduling (Healthcare ICT)

By analyzing historical patient admission patterns (e.g., more emergency admissions on weekends, more cardiac events in winter), hospitals can predict when and which departments will face high demand. ICT systems automatically adjust scheduling, ensuring enough doctors, nurses, and medical equipment are available — reducing patient wait times and staff overtime costs.

Q6(c) — Distinguish between Correlation and Regression Analysis. Applications of each. [2 Marks]

Dimension	Correlation	Regression
Definition	Measures the strength and direction of relationship between two variables	Models the relationship to predict the value of one variable based on another
Output	Correlation coefficient (r), range: -1 to +1	Regression equation: $Y = a + bX$

Direction	Bidirectional — X relates to Y is same as Y relates to X	Directional — X predicts Y; roles are not interchangeable
Prediction	Cannot make predictions	Primary purpose is prediction
Example	Study hours and exam scores are positively correlated ($r = 0.85$)	$Y = 40 + 6X$ predicts sales (Y) from advertising spend (X)

Applications of Correlation:

- Finance: Measuring correlation between stocks to build diversified portfolios
- Medical Research: Assessing whether smoking level correlates with lung capacity

Applications of Regression:

- Sales forecasting based on advertising spend
- Predicting house prices from square footage, location, and age

Q6(d) — KNN as 'lazy learner'. Advantages and disadvantages. [2 Marks]

KNN is called a 'lazy learner' because it does NOT build an explicit model during training. Instead, it simply stores the entire training dataset. When a new query arrives, it computes distances to all stored points at that moment to make a prediction. This is in contrast to 'eager learners' (like Decision Trees) which build a model during training and use it later.

Advantages of the Lazy Learning Property:

- No training time — predictions can begin immediately after data is stored.
- Naturally adapts to new data — adding training samples improves performance without retraining.
- Works well with non-linear decision boundaries.

Disadvantages:

- Very slow at prediction time — must calculate distance to ALL training points for each query.
- High memory usage — the entire training dataset must be stored.
- Sensitive to irrelevant features and different scales — feature scaling (normalization) is required.

Q6(e) — How visualization choice impacts dataset understanding. Wrong choice example. [2 Marks]

Chart Type	Best Used For	Example
Scatter Plot	Showing relationship between two continuous variables	Visualizing correlation between advertising spend and sales
Histogram	Showing distribution/frequency of a single continuous variable	Visualizing distribution of student exam scores
Box Plot	Showing spread, median, quartiles, and outliers in a dataset	Comparing salary distribution across departments
Heatmap	Showing correlation between multiple variables or intensity across a matrix	Visualizing correlation matrix of all features in a dataset

Wrong Choice Example — Misleading Interpretation:

Scenario: A company wants to show that sales have been consistently growing over 12 months. If they use a Histogram instead of a Line Graph, viewers will see the frequency distribution of monthly sales values — not the trend over time. The histogram might show that most months had moderate sales (a

central peak), giving the false impression of stable, flat performance when in reality there was strong upward momentum. The time dimension is completely lost.

Question 7

Q7(a) — SWOT Analysis of an ICT cloud-based education platform in emerging markets. [3 Marks]

Factor	Points
Strengths (S)	Low marginal cost of digital content delivery; scalable infrastructure; accessible on mobile devices widely used in emerging markets; ability to offer affordable pricing compared to physical institutions
Weaknesses (W)	Dependency on reliable internet connectivity (low in rural areas); difficulty in establishing brand trust in new markets; high initial investment in content development and localization
Opportunities (O)	Large young population with growing internet penetration; government digitization initiatives; demand for affordable skill-building; potential partnerships with local institutions
Threats (T)	Competition from established global platforms (Coursera, Khan Academy); data privacy regulatory differences across countries; currency fluctuation affecting pricing; digital divide limiting addressable market

Activity Network in ICT Business: An activity network (also called a Project Network or CPM diagram) maps all tasks, their sequence, dependencies, and durations in a project. For an ICT platform launch, nodes represent activities (e.g., platform development, content creation, regulatory approval, marketing), and arrows show dependencies. The Critical Path identifies the longest sequence — the minimum time needed to launch.

Q7(b) — KNN (K=3) to classify new sample (Weight=160, Taste=6). [4 Marks]

Dataset:

Weight	Taste	Fruit
150	8	Coconut
130	6	Coconut
200	4	Pineapple
180	5	Pineapple

Step 1: Calculate Euclidean Distance from (160, 6):

Point	Weight	Taste	Calculation	Distance	Fruit
P1	150	8	$\sqrt{[(160-150)^2 + (6-8)^2]} = \sqrt{100+4}$	$\sqrt{104} \approx 10.20$	Coconut
P2	130	6	$\sqrt{[(160-130)^2 + (6-6)^2]} = \sqrt{900+0}$	$\sqrt{900} = 30.00$	Coconut
P3	200	4	$\sqrt{[(160-200)^2 + (6-4)^2]} = \sqrt{1600+4}$	$\sqrt{1604} \approx 40.05$	Pineapple

P4	180	5	$\sqrt{[(160-180)^2+(6-5)^2]} = \sqrt{400+1}$	$\sqrt{401} \approx 20.02$	Pineapple
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Step 2: K=3 Nearest Neighbors:

P1 (10.20, Coconut), P4 (20.02, Pineapple), P2 (30.00, Coconut)

Step 3: Majority Vote:

Coconut: P1, P2 = 2 votes | Pineapple: P4 = 1 vote

Classification Result: COCONUT — The new sample (Weight=160g, Taste=6) is classified as Coconut based on K=3 nearest neighbor majority vote.

Q7(c) — Explain Positive, Negative, and No Correlation with scatter plots. Linear vs Non-linear. [3 Marks]

Types of Correlation:

Type	Description	Example	Scatter Plot Pattern
Positive Correlation	Both variables increase or decrease together; move in same direction	Study hours and exam score — more study leads to higher marks	Dots trend upward from left to right (r close to +1)
Negative Correlation	One variable increases while the other decreases; move in opposite directions	Price of a product and quantity demanded — higher price, fewer buyers	Dots trend downward from left to right (r close to -1)
No Correlation	No consistent relationship between variables	Shoe size and IQ score — completely unrelated variables	Dots scattered randomly with no directional pattern (r close to 0)

Linear vs Non-linear Correlation:

Dimension	Linear Correlation	Non-linear Correlation
Pattern	Relationship follows a straight line	Relationship follows a curve (quadratic, exponential, etc.)
Rate of Change	Constant — for every unit increase in X, Y changes by the same amount	Variable — the change in Y per unit of X is not constant
Coefficient	Pearson's r measures strength well	Pearson's r may be misleading; Spearman's rank correlation is better
Example	Height vs weight (roughly linear)	Income vs spending — as income rises, spending rises fast at first then plateaus

Q7(d) — Regression process in data analysis. Characteristics of linear regression. Business example. [3 Marks]

The Regression Process:

1. Define the dependent variable (Y) to be predicted and independent variable(s) (X) as predictors.
2. Visualize the relationship using a scatter plot to check if a linear pattern exists.

3. Fit the regression line: $Y = a + bX$, where a = intercept (Y when $X=0$) and b = slope (rate of change).
4. Evaluate model fit using R^2 (coefficient of determination), RMSE, and MAE.
5. Use the validated equation to predict Y for new values of X .

Key Characteristics of Linear Regression:

- Linearity: Assumes a straight-line relationship between X and Y .
- Independent Errors: Residuals (actual - predicted) must be random, not show a pattern.
- Normality: Residuals should be normally distributed.
- Homoscedasticity: Variance of errors should be constant across all values of X .
- No Multicollinearity (Multiple Regression): Independent variables must not be highly correlated with each other.

Business Example:

A clothing retailer wants to predict monthly revenue (Y) based on marketing budget (X). After collecting 24 months of data, the fitted equation is $Y = 150,000 + 8.5X$. This means for every additional 1,000 BDT spent on marketing, revenue increases by 8,500 BDT. The intercept 150,000 represents baseline revenue even without marketing. Using this model, the company can determine the optimal marketing budget and set realistic revenue targets for the next quarter.

TUTORIAL QUESTIONS — WITH ANSWERS

Topic 01: Data Analytics Fundamentals

Section A: Short Answer Questions

T1 — Define Data Analytics. How is it different from Data Analysis?

Dimension	Data Analytics	Data Analysis
Focus	Broad: entire process from collection to insight	Narrower: specifically the examination/interpretation step
Scope	Includes strategy, tools, and business application	Statistical and mathematical examination of data
Output	Business insights and recommendations	Patterns, summaries, conclusions
Relationship	Encompasses data analysis as one of its steps	A subset of the data analytics process

T2 — List and explain the 5 key steps in the Data Analysis Process.

- Step 1 — Define the Question: Identify the business problem and frame a specific, answerable question.
- Step 2 — Collect the Data: Gather relevant data from first-party, second-party, or third-party sources.
- Step 3 — Clean the Data: Remove errors, duplicates, and outliers; handle missing values; standardize formats.
- Step 4 — Analyze the Data: Apply descriptive, diagnostic, predictive, or prescriptive techniques to extract insights.

- Step 5 — Share Results: Present findings using visualizations and clear communication to stakeholders.

T3 — What is an outlier? How does it affect mean, median, and standard deviation?

An outlier is a data point that falls far outside the typical range of values in a dataset — significantly higher or lower than most other values.

- Mean: Severely affected — even a single extreme outlier pulls the mean toward it, making it unrepresentative.
- Median: Largely unaffected — as the middle value of sorted data, it is resistant to extreme outliers.
- Standard Deviation: Significantly inflated — since SD measures spread around the mean, outliers that distort the mean also inflate the SD, making data appear more variable than it truly is.

T4 — Distinguish First-Party, Second-Party, and Third-Party data.

Type	Definition	Example
First-Party	Data collected directly by your own organization from your own customers or operations	Bkash transaction history, your company's CRM data, website analytics
Second-Party	First-party data of another organization, shared or purchased through a direct partnership	A shipping company sharing delivery data with an e-commerce partner
Third-Party	Data aggregated from many sources by an outside organization and sold/provided to others	Kaggle datasets, government portals like data.gov, market research reports

T5 — What are the 4 types of Data Analysis? Which is the most complex and why?

Type	Question Answered	Description
Descriptive	What happened?	Summarizes historical data to reveal past events and trends
Diagnostic	Why did it happen?	Identifies causes and root reasons behind observed events
Predictive	What will happen?	Uses historical patterns and ML to forecast future outcomes
Prescriptive	What should we do?	Recommends optimal actions by combining all other types

Prescriptive analytics is the most complex because it incorporates outputs from descriptive, diagnostic, and predictive analytics simultaneously, then applies optimization algorithms or simulation to determine the BEST possible course of action. It requires the most data, computing power, and domain expertise.

T6 — What is Data Quality? Name the 6 dimensions of data quality.

Data Quality measures how well a dataset is suited for its intended purpose — how accurate, complete, and reliable it is for the task at hand. The 6 dimensions are: Accuracy (is the data correct?), Completeness (is all needed data present?), Consistency (does data match across all systems?), Validity (does data follow required formats/rules?), Uniformity (are all values in the same unit/format?), and Relevance/Timeliness (is the data current and applicable?).

T7 — Explain the 8 steps of Data Cleaning with brief examples.

6. Remove unwanted observations — Delete irrelevant rows. E.g., for vegetarian diet analysis, remove all meat-related records.
7. Fix structural errors — Standardize typos. E.g., 'Iron' and 'iron' standardized to one form.
8. Standardize your data — Use consistent units and formats. E.g., all dates in DD/MM/YYYY format.
9. Remove unwanted outliers — Only remove outliers proven to be errors. E.g., a test score of 200/100 is physically impossible.
10. Fix contradictory data errors — Resolve inconsistencies. E.g., race total time not matching sum of lap times.
11. Fix type conversion errors — Ensure correct data types. E.g., storing ages as numbers, not text strings.
12. Deal with missing data — Remove, impute, or flag missing values based on context.
13. Validate your dataset — Final review to confirm all cleaning steps are complete and data is analysis-ready.

T8 — Difference between Quantitative and Qualitative data. 2 examples each.

Dimension	Quantitative (Numeric)	Qualitative (Descriptive)
Definition	Data expressed as numbers that can be measured and compared mathematically	Data expressed as descriptive text, categories, or labels — not numeric
Example 1	Monthly sales: 10,000 BDT; Number of website visits: 5,000	Customer reviews: 'Excellent service, very fast delivery'
Example 2	Student's exam score: 87 out of 100	Survey response: 'Strongly Agree' or 'Disagree'
Analysis	Mean, median, standard deviation, regression	Thematic analysis, word frequency, sentiment analysis

Section B: Calculation-Based Questions

T9 — Find Mean, Median, Mode, Variance, and SD for: {4, 8, 6, 5, 3, 2, 8, 9, 2, 5}

Sorted: 2, 2, 3, 4, 5, 5, 6, 8, 8, 9 (n = 10)

Mean = $(2+2+3+4+5+5+6+8+8+9) / 10 = 52 / 10 = 5.2$

Median: n=10 (even), average of 5th and 6th terms = $(5+5)/2 = 5.0$

Mode: 2, 5, and 8 each appear twice — all three are modes (multimodal dataset)

Variance: Deviations from mean (5.2): $(2-5.2)^2=10.24$, $(2-5.2)^2=10.24$, $(3-5.2)^2=4.84$, $(4-5.2)^2=1.44$, $(5-5.2)^2=0.04$, $(5-5.2)^2=0.04$, $(6-5.2)^2=0.64$, $(8-5.2)^2=7.84$, $(8-5.2)^2=7.84$, $(9-5.2)^2=14.44$

Sum of squared deviations = 57.6

Variance = $57.6 / 10 = 5.76$

Standard Deviation = $\sqrt{5.76} = 2.4$

Mean = 5.2, Median = 5.0, Mode = 2, 5, 8 | Variance = 5.76, SD = 2.4

T10 — Calculate Range and Coefficient of Range for: {45, 55, 63, 76, 67, 84, 75, 48, 62, 65}

$X_{\text{max}} = 84$, $X_{\text{min}} = 45$

Range = $84 - 45 = 39$

Coefficient of Range = $(84 - 45) / (84 + 45) = 39 / 129 \approx 0.302$

Range = 39, Coefficient of Range ≈ 0.302

T11 — Mean of 14 observations is 17. Find missing observation x.

Known values: 26, 12, 14, 15, x, 17, 9, 11, 18, 16, 28, 20, 22, 8

Sum of known values = $26+12+14+15+17+9+11+18+16+28+20+22+8 = 216$

Mean formula: $17 = (216 + x) / 14 \rightarrow 17 \times 14 = 216 + x \rightarrow 238 = 216 + x$

$x = 238 - 216 = 22$

Missing observation x = 22

T12 — For {19, 21, 23, 20, 23, 27, 25, 24, 31}: Find Q1, Q2 (Median), Q3, and IQR.

Sorted: 19, 20, 21, 23, 23, 24, 25, 27, 31 (n = 9)

Q2 (Median) = $(9+1)/2 = 5$ th term = 23

Q1 = $(9+1)/4 = 2.5$ th term = average of 2nd and 3rd = $(20+21)/2 = 20.5$

Q3 = $3(9+1)/4 = 7.5$ th term = average of 7th and 8th = $(25+27)/2 = 26$

IQR = $Q3 - Q1 = 26 - 20.5 = 5.5$

Q1 = 20.5, Q2 (Median) = 23, Q3 = 26, IQR = 5.5

Section C: Applied Questions

T13 — $Y = 50 + 8X$. Predict sales when $X = 5000$ BDT. What does intercept (50) mean?

$Y = 50 + 8(5) = 50 + 40 = 90$ units [X is in thousands, so 5000 BDT = $X = 5$]

Intercept (50): Even without any advertising spend ($X = 0$), the model predicts 50 units will be sold. This represents the baseline sales driven by existing brand loyalty, returning customers, or organic demand independent of advertising efforts.

T14 — Real-life Predictive Analytics scenario from daily life.

Scenario — Ride-hailing Surge Pricing (Pathao/Uber):

- Data Collected: Current ride requests, available drivers, time of day, day of week, weather conditions, historical demand patterns for that area and time.
- Predictive Model: The algorithm predicts demand in the next 15–30 minutes based on historical patterns. For example, it detects that demand always surges in Dhanmondi between 8–9 AM on weekdays.
- Decision Made: The app automatically applies surge pricing to attract more drivers to high-demand areas, and nudges nearby idle drivers to reposition. Passengers are shown estimated wait times. The platform balances supply and demand in real time — replacing manual dispatcher decisions with automated, data-driven responses.

T15 — Preprocessing steps for PQS.csv before building an ML model.

- Handle Missing Values: Identify missing entries in numerical columns (impute with mean/median) and categorical columns (impute with mode or create an 'Unknown' category).
- Remove Duplicates: Check for identical rows and remove them to prevent the model from over-learning repeated examples.
- Outlier Treatment: Use IQR or Z-score to identify outliers in numerical features; investigate and decide to remove or cap (winsorize) extreme values.
- Encode Categorical Variables: Convert text categories to numbers using Label Encoding (for ordinal data) or One-Hot Encoding (for nominal data like color, city, gender).
- Feature Scaling: Apply Min-Max Normalization or Z-score Standardization to numerical features so that no variable dominates due to its scale (critical for KNN, SVM, neural networks).
- Feature Selection: Remove irrelevant or redundant columns that do not contribute to predicting the target variable. Use correlation matrices or feature importance scores.
- Train-Test Split: Divide the dataset (e.g., 80% train, 20% test) before fitting the model to ensure evaluation on unseen data.

Topic 02: Statistics and Business Intelligence

Section A: Conceptual Questions

T16 — What is a Normal Distribution? State 3 key properties.

A Normal Distribution (also called a Bell Curve) is a symmetric, continuous probability distribution where data clusters around the mean, with progressively fewer observations further from the center. It is the most common distribution in natural and social phenomena.

3 Key Properties:

- Symmetry: The distribution is perfectly symmetric around the mean — the left side mirrors the right. Mean = Median = Mode, all at the same central point.
- 68-95-99.7 Rule: Approximately 68% of data falls within 1 SD of the mean, 95% within 2 SDs, and 99.7% within 3 SDs. This makes standard deviation extremely meaningful for interpretation.
- Asymptotic Tails: The curve approaches but never touches the X-axis, meaning theoretically all values are possible, but extreme values are extremely rare.

T17 — Left-Skewed (Negative Skew) distribution and its effect on Mean vs Median.

A left-skewed (negatively skewed) distribution has a long tail extending to the LEFT. Most data points are concentrated on the RIGHT side (higher values), with a few very low values pulling the tail leftward. The peak (mode) is on the right.

Effect on Mean vs Median: The few very low extreme values drag the mean toward the left. As a result, the Mean is LOWER than the Median, and the Median is lower than the Mode.

Order: Mean < Median < Mode

Example: If most students score 80–95 on a test but a few score very low (10–20), the distribution is left-skewed. The mean might be 75 (dragged down by low scores) while the median is 85 (unaffected by extremes).

T18 — Positive, Negative, and No Correlation with examples and scatter plot descriptions.

Type	Description	r Value Range	Scatter Plot	Example
Positive	Both variables move in same direction — when X rises, Y rises	0 to +1	Dots trend upward from bottom-left to top-right	More study hours → higher exam score
Negative	Variables move in opposite directions — when X rises, Y falls	-1 to 0	Dots trend downward from top-left to bottom-right	Higher product price → lower quantity sold
No Correlation	No consistent relationship between variables	Near 0	Dots scattered randomly with no pattern	Shoe size vs intelligence test score

T19 — What is Business Intelligence (BI)? Complete the flow: Data → ? → Decision → Action.

Business Intelligence (BI) is a comprehensive set of strategies, technologies, and tools that organizations use to collect, integrate, analyze, and present business data to support smarter decision-making.

Complete BI Flow: Data → Information → Decision → Action

- Data: Raw, unprocessed facts from operational systems (sales figures, transactions, customer records).

- **Information:** Data that has been processed, cleaned, and contextualized to become meaningful (monthly revenue trend, customer retention rate).
- **Decision:** Leaders use the information to choose a course of action (launch a discount campaign, hire more staff).
- **Action:** The decision is implemented and results are measured, generating new data that feeds back into the cycle.

T20 — Describe the evolution of analytics from 1970s to 2010s Big Data era.

Decade	Key Technology	Primary Innovation
1970s	DSS, Expert Systems, AI basics	First computer systems to assist managerial decision-making with structured data
1980s	Relational DBMS, ERP, On-Demand Reporting	Relational databases organized business data; ERPs integrated cross-functional operations
1990s	Data Warehousing, Dashboards, EIS/ESS	Separate analytical stores emerged; executives could see real-time summaries and KPIs
2000s	BI Platforms, Data Mining, SaaS, Cloud	BI matured into enterprise-wide self-service platforms; cloud made tools broadly accessible
2010s	Big Data, Hadoop, NoSQL, Social Analytics	Internet and mobile explosion created vast unstructured data requiring new processing tools

T21 — What are the 4 Vs of Big Data? Explain each with a practical example.

V	Definition	Practical Example
Volume	Enormous quantity of data beyond traditional system capacity	Walmart processes over 1 million customer transactions per hour
Velocity	Speed at which data is generated and must be processed	Stock market tickers generate thousands of data points per second; delays cost money
Variety	Multiple formats: structured, unstructured, semi-structured	A hospital collects numeric lab results, X-ray images, doctor's handwritten notes, and audio recordings
Veracity	Uncertainty and trustworthiness of data quality	Social media data contains sarcasm, fake accounts, and errors that reduce analytical reliability

T22 — What is OLAP? How does it differ from OLTP? 2 examples of each.

Dimension	OLAP	OLTP
Full Name	Online Analytical Processing	Online Transaction Processing
Purpose	Complex data analysis, reporting, and business intelligence	Day-to-day operational transaction management
Data Volume	Large — historical data spanning months or years	Current data — recent transactions
Query Type	Complex, multi-dimensional analytical queries	Simple, fast create/read/update/delete operations

Response Time	Seconds to minutes (complex aggregation)	Milliseconds (simple lookups)
Example 1	Analyzing 3 years of sales by region, product, and quarter	Processing a customer's online payment
Example 2	Generating a monthly profit-and-loss comparison across departments	Updating inventory when a product is purchased

T23 — Define Analytics Ecosystem. Name and explain any 5 players.

The Analytics Ecosystem is the complete network of interdependent organizations, institutions, and tools that collaborate to enable data-driven decision-making for analytics user organizations.

Player	Role in Ecosystem
Analytics User Organization	The central business entity that uses analytics to inform strategy, operations, and decisions
Data Generation Infrastructure Providers	Supply sensors, mobile devices, web platforms, and IoT systems that produce raw data
Data Warehouse Providers	Offer storage and management systems for large-scale cleaned and organized analytical data (e.g., Amazon Redshift, Snowflake)
Analytics-Focused Software Developers	Build specialized BI and analytical tools like Tableau, Power BI, SPSS, SAS
Regulators and Policy Makers	Governments and agencies that establish legal frameworks governing data collection, usage, privacy, and security

Section B: Application Questions

T24 — Uniform distribution: fair die (1 to 6). Probability of a single outcome. [Graph description.]

For a uniform distribution with $a = 1$ (minimum) and $b = 6$ (maximum):

$f(x) = 1 / (b - a) = 1 / (6 - 1) = 1/5 = 0.2$ per unit — but since there are 6 discrete outcomes: $P(\text{any single outcome}) = 1/6 \approx 0.167$ (16.7%)

Graph Description: A bar chart with 6 equally-spaced bars (one for each die face: 1, 2, 3, 4, 5, 6) on the X-axis, with the Y-axis showing probability. All bars are exactly the same height at 0.167, forming a flat, rectangular distribution. No value has higher or lower probability than another — perfectly uniform.

T25 — Linear vs Non-linear Correlation. Real-life non-linear example.

Dimension	Linear Correlation	Non-linear Correlation
Shape	Straight-line relationship on scatter plot	Curved relationship (parabola, exponential, etc.)
Rate of Change	Constant — each unit increase in X produces the same change in Y	Variable — change in Y varies at different values of X
Pearson's r	Accurately captures the relationship	May underestimate; Spearman's or non-linear methods better
Example	Hours studied and marks scored (roughly constant return per hour)	Medicine dosage and effectiveness — low dose: little effect; optimal dose: high effect; overdose: harm

Real-life Non-linear Example: Temperature and ice cream sales. From 0°C to 20°C, sales grow slowly. From 20°C to 35°C, sales accelerate rapidly. Above 40°C, sales may drop (too hot to go out). This S-shaped or curve relationship cannot be captured by a straight line.

T26 — ETL in a BI system: Extract, Transform, Load explained in a business scenario.

Business Scenario: A retail chain integrating sales data from physical stores and an online platform.

Extract: Pull raw transaction records from the physical store's POS (Point of Sale) system and simultaneously extract online orders from the e-commerce platform's database. Data arrives in different formats — the POS exports CSV files while the online platform provides JSON via API.

Transform: Standardize date formats (all to YYYY-MM-DD), convert currencies to the same unit, remove duplicate transactions, map product codes that differ between systems, handle null values in discount fields, and classify transactions by region and product category.

Load: Insert the clean, standardized, unified dataset into the central data warehouse. The data is now organized in subject-oriented tables (Sales, Products, Customers) ready for analysts to query with BI tools like Tableau or Power BI.

T27 — How do Data Marts differ from a Data Warehouse? Why create a Finance Data Mart?

Dimension	Data Warehouse	Data Mart
Scope	Enterprise-wide; contains data for ALL departments	Department-specific; contains only relevant subject data
Size	Very large — terabytes to petabytes	Smaller — gigabytes to terabytes
Users	All business analysts and BI tools across organization	Specific team (Finance, Marketing, Sales)
Creation	Built first; data marts are derived from it	Built from the warehouse or directly from sources
Access Speed	Slower due to large volume	Faster due to smaller, focused dataset

Why a Finance Data Mart? A separate Finance Data Mart is created so the finance team can query only their relevant data (P&L statements, budget vs actuals, accounts receivable/payable, tax records) without waiting for queries to scan the entire enterprise warehouse. It improves query speed, enforces data security (only finance staff have access), and allows finance-specific data models and calculations (e.g., fiscal year definitions, currency hedging) without affecting other departments.

Section C: Mixed / Higher-Order Thinking

T28 — Is a Descriptive → Predictive → Prescriptive hierarchy a good idea? Argue with examples.

YES — following this analytics hierarchy is strongly advisable, both logically and practically.

Arguments in Favor:

- Foundation First: Descriptive analytics answers 'what happened' by summarizing historical data. Without this understanding, you cannot meaningfully identify which variables to use in predictive models.
- Causal Understanding: Diagnostic analytics (a subset of descriptive) answers 'why it happened.' Knowing WHY events occur is essential for building accurate predictive models — otherwise, you may be predicting outcomes from the wrong drivers.

- **Reliable Forecasting:** Predictive analytics answers 'what will happen' using historical patterns. These forecasts must be reliable before prescriptive analytics can recommend actions based on them.
- **Informed Action:** Prescriptive analytics answers 'what should we do' by evaluating scenarios based on predictions. If predictions are unreliable (built without proper descriptive foundations), prescriptive recommendations will also be unreliable.

Example:

A bank wants to reduce loan defaults. First, descriptive analytics reveals that default rates are highest in specific age groups and loan sizes. Diagnostic analytics shows that these customers typically have high debt-to-income ratios. Predictive analytics then forecasts which current applicants are likely to default. Finally, prescriptive analytics recommends: 'reject applications with a predicted default probability > 30%, and offer debt counseling to those between 20–30%.' Skipping descriptive/diagnostic would mean the predictive model has no validated features to work with.

T29 — How can analytics aid objective decision making? Business example where subjective replaced by data-driven.

Analytics replaces subjective human judgment — which is subject to cognitive biases, incomplete information, and emotional influence — with evidence-based reasoning derived from systematic data examination.

How Analytics Ensures Objectivity:

- Removes personal bias by relying on patterns in large datasets rather than individual opinion or experience.
- Enables consistent decision-making — the same model applied to the same data always produces the same recommendation.
- Quantifies uncertainty — instead of gut feelings, decision-makers receive probability scores (e.g., '72% likelihood of customer churn').
- Tracks outcomes over time and refines models, creating a learning feedback loop that improves future decisions.

Business Example — HR Recruitment:

Previously, hiring managers subjectively evaluated resumes based on gut feeling, university prestige, or personal rapport — introducing unconscious bias and inconsistency. After implementing an analytics-driven candidate scoring system that evaluates candidates on quantifiable factors (GPA, relevant experience months, technical test scores, role-specific competency metrics), the company reduced bias complaints by 40%, improved 1-year retention of new hires by 25%, and reduced time-to-hire by 30% — all because data replaced subjective judgment.

T30 — Compare Descriptive and Diagnostic Analytics. Which to use first for a sales drop? Why?

Dimension	Descriptive Analytics	Diagnostic Analytics
Question Answered	What happened?	Why did it happen?
Focus	Summarizing historical facts	Finding causes and root reasons
Techniques	Aggregation, data mining, dashboards	Drill-down analysis, correlation, segmentation
Output	KPIs, trend charts, summary statistics	Root cause identification, contributing factors
Example	Sales dropped 20% in Q3 2023	Q3 drop caused by losing 3 major retail clients due to poor after-sales support

Which to Use First — and Why?

For a company whose sales dropped last quarter, DESCRIPTIVE analytics should be applied FIRST. Before investigating why sales dropped, you must first confirm and characterize what happened precisely: By how much did sales drop? In which regions, products, or customer segments was the drop concentrated? Was it a gradual decline or sudden? Which months were worst?

Without this descriptive foundation, any diagnostic investigation is unfocused — you would be searching for causes in the wrong places. Once descriptive analytics has mapped out the 'shape' of the problem (sales of Product A dropped 35% in the northern region among B2B clients in September), diagnostic analytics can efficiently narrow down to probable causes (was there a competitor promotion? a supply chain issue? a loss of a key account?). The two types work in sequence, not in isolation.

— End of Answer Booklet —

Prepared with full answers based on Topic-01 and Topic-02 Study Notes